Lab: Neural Oscillations and Predictive Coding

> **Learning Goal:** Analyze oscillatory evidence for predictive coding and attention as precision optimization.

Part 1: Oscillatory Predictive Coding Mapping

Exercise: Complete the comprehensive mapping:

Frequency Band	Frequency Range	Cortical Layers	Direction	Computational Role	Key Evidence
Gamma	30-100 Hz	L2/3 (superficial)	Feedforward (bottom-up)	Prediction errors	MMN, mismatch responses, stimulus-driven increases
Beta	13-30 Hz	L5/6 (deep)	Feedback (top-down)	Predictions	Increases with expected stimuli, motor preparation
Alpha	8-12 Hz	Variable (thalamo-cortical)	Modulatory	Precision gating (attention)	Decreases in attended areas, increases in suppressed areas
Theta	4-8 Hz	Hippocampus, frontal	Long-range coordination	Temporal organization, working memory	Phase-amplitude coupling with gamma
Delta	1-4 Hz	Deep layers	Temporal prediction	Entrainment to rhythmic stimuli	Speech processing, temporal attention

Now explain: Why does the brain need multiple frequency bands? What would happen if all prediction errors were transmitted at the same frequency?

Write your response here

Part 2: Mismatch Negativity Analysis

> **Learning Goal:** Analyze the MMN as a neural prediction error signal.

Exercise: Design a complete MMN experiment:

1. **Stimulus sequence**: Define your standard (repeated) and deviant (rare) stimuli. Be specific (e.g., 1000 Hz standard tone, 1500 Hz deviant, 80/20 ratio, 500ms SOA).
2. **Predicted neural response**: What ERP component do you expect? At what latency? Over which electrodes?
3. **Active Inference interpretation**: Explain the MMN as a prediction error — what model generates the prediction? What is the error signal?
4. **Pharmacological manipulation**: If you administered ketamine (NMDA antagonist), how would the MMN change? Why?
5. **Clinical application**: How does MMN change in schizophrenia? What does this suggest about predictive processing deficits?

Write your response here

Part 3: Attention as Precision

> **Learning Goal:** Analyze attention experiments through the precision framework.

Exercise: For each attention paradigm, explain the precision mechanism:

Posner cueing task: A cue directs attention to a spatial location. Targets appearing at the cued location are detected faster.

- What happens to alpha over the attended hemifield?
- How does this implement precision weighting?
- What happens on invalid trials (target at uncued location)?

Visual search: Finding a target among distractors (e.g., red T among green T's).

- How do frontal theta oscillations represent the search template?
- How does this template modulate visual gamma (feature-level processing)?
- How does this implement top-down precision allocation?

Attentional blink: Missing the second of two targets presented in rapid succession.

- How does the first target deplete precision resources?
- What oscillatory signature marks the "blink" period?
- How can the blink be reduced (e.g., through meditation training)?

Write your response here

Part 4: Sensory Attenuation Experiment

> **Learning Goal:** Analyze self-generated vs. external stimulation through predictive coding.

Exercise: Consider the tickling phenomenon — you cannot tickle yourself.

1. Draw the active inference circuit: motor command → predicted sensory consequence → actual sensory consequence → prediction error
2. Why is self-generated touch attenuated? (The motor system predicts the touch; prediction error is small)
3. How does the cerebellum contribute to the forward model?

4. Why can schizophrenia patients sometimes tickle themselves? (Disrupted forward model → prediction fails)
5. Design an experiment to measure sensory attenuation neurally. What oscillatory signature would you expect?

Write your response here

Part 5: Reflection

In 300 words, reflect: The oscillatory predictive coding framework provides an elegant mapping between computational theory and neural implementation. But oscillations are noisy, variable, and hard to measure. How confident should we be that these frequency bands really map onto predictions and errors, rather than reflecting other computational processes?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Systems mapping	Oscillatory frequency roles
2	Experiment design	MMN as prediction error
3	Paradigm analysis	Attention as precision
4	Circuit analysis	Sensory attenuation
5	Critical evaluation	Confidence in the framework